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3.1.12

Alg. MidTerm Redo

4b. $\sum_{i=1}^{n} 4^i = \sum_{i=0}^{n} 4^i - \sum_{i=0}^{0} 4^i = 4^{n+1} - 1 - 1 = 4^{n+1} - 2$

4a. $\sum_{i=1}^{n} 2i - \sum_{i=1}^{2} 2i = 2\left(\frac{n(n+1)}{2}\right) - 6$

3b. $2^{n-1} + \sum_{i=0}^{n-1} 2^i = 2^{n-1} + 2^{n-1} - 1 = 2^n - 1$

11.b
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Multiply (m,n)
    if (n==1)
        return n;
    else
        return 2 * multiply (m, n/2);
```

11a. Use a recursive function that can run in faster than linear time.

10.a - Marked F6 as wrong.. $3^n = \Omega(n)$ correct!

a - for $(\lg n)^2 \neq \Omega(n)$

$(\lg 32)^2 \neq \Omega(32)$

$5^2 \neq \Omega\, 32$

a - for F4 - $\lg(n^2) \neq \Omega(n)$

$\lg 16^2 \neq \Omega(16)$

$8 \neq \Omega(16)$

a - Marked F4 as correct. $\sqrt{n} \neq \Omega(n)$

$\sqrt{16} \neq \Omega(16)$

b - $3^n \neq O(n^2)$

$3^4 \neq O(4^2)$

b - F7 is correct because Big-O is less than or equal to, so $n^2 = O(n^2)$

10 c – Marked F_5 as in $\Theta(\lg n)$...

$\lg(n^2) \neq \Theta(\lg n)$

$\lg(8^2) \neq \Theta(\lg 8)$

$\lg(64) \neq \Theta(\lg 8)$

d. The rate that each function grows is exponential, so both are in the same theta class.

6. $T(n) = \Theta(n \lg n)$

7b. $T(n) = \Theta(n \lg n)$

3a. $T(n) = T(n-2)+1+1 = T(n-3)+1+1+1 = T(n-4)+1+1+1+1$

$\sum_{i=1}^{n} 1$

n